

CONTROLLING A ROBOTIC CAR Through MATLAB GUI

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This project presents the implementation of a MATLAB based graphical user interface (GUI) to control the movement of a robotic car. The commands to move the car in forward, reverse, right and left directions are sent from the GUI and processed by Arduino Uno board mounted on top of the car. Authors' prototype is shown in Fig. 1.

Circuit and working

In this project, MATLAB based GUI communicates with Arduino through a USB data cable. You can send commands to Arduino to move the car in a specific direction by pressing the corresponding push button in the GUI. The particular push button, when pressed, executes callback function corresponding to that push button in the MATLAB program, car.m. Within that function, instructions to set/reset the digital I/O pins (10, 11, 12 and 13) of Arduino are executed to drive the car in the desired direction.

Block and circuit diagrams for controlling the robot are shown in Figs 2 and 3, respectively.

Arduino Uno board. Arduino Uno is an AVR ATmega328P microcontroller (MCU) based development board with six analogue input pins and 14 digital I/O pins. The MCU has 32kB of ISP flash memory, 2kB SRAM and 1kB EEPROM. The board provides serial communication capability via UART, SPI and I2C. The MCU can operate at a clock frequency of 16MHz. In this project, digital I/O pins 10, 11, 12 and 13 of Arduino are configured as output pins.

L293D IC. This is a dual H-Bridge high-current motor driver IC. This



Fig. 1: Author's prototype of the MATLAB GUI based robotic car

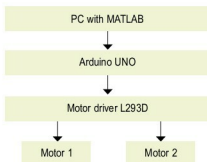


Fig. 2: Block diagram for controlling the robotic car through MATLAB based GUI

TABLE I SIGNAL BITS SENT FROM ARDUINO TO L293D		
Command from MATLAB	Arduino output pins (10, 11, 12 and 13)	L293D output pins
Forward	LHLH	LHLH
Reverse	HLHL	HLHL
Right	LLLH	LLLH
Left	LHLL	LHLL
Stop	LLLL	LLLL

IC is used since the digital pins of Arduino Uno cannot source enough current to drive the motors of the robotic car. H-bridges are also useful in controlling the direction of rotation of a motor. Enable pins (pins 1

and 9) of the IC being active high are connected to 9V.

Four input pins IN1, IN2, IN3 and IN4 of L293D have been connected to digital I/O pins 10, 11, 12 and 13, respectively, of Arduino. Four output pins (OUT1, OUT2, OUT3 and OUT4) of the IC are connected to the

two motors (M1 and M2) with their terminals marked as A, B, C and D, respectively, as shown in Fig. 3.

On pressing a particular push-button in the GUI, digital signals sent from Arduino to L293D are listed in Table I.

Software program

Circuit operation is performed using the software program loaded into the internal memory of Arduino board. The program implements all required functionalities including user inputs through MATLAB GUI interface. The steps are listed below:

1. Arduino IDE 1.6.5 is used for programming Arduino Uno. The latest IDE can be downloaded for free from Arduino's official website. After downloading Arduino IDE, install it. Note the directory of Arduino IDE installation.

2. Download Legacy MATLAB and Simulink Support for Arduino package from MathWorks website. Extract the compressed folder named ArduinoIO. From this folder, copy pde folder and paste it in C:\Program Files (or Program files x86)\Arduino\Libraries (path may be different depending on the instal-